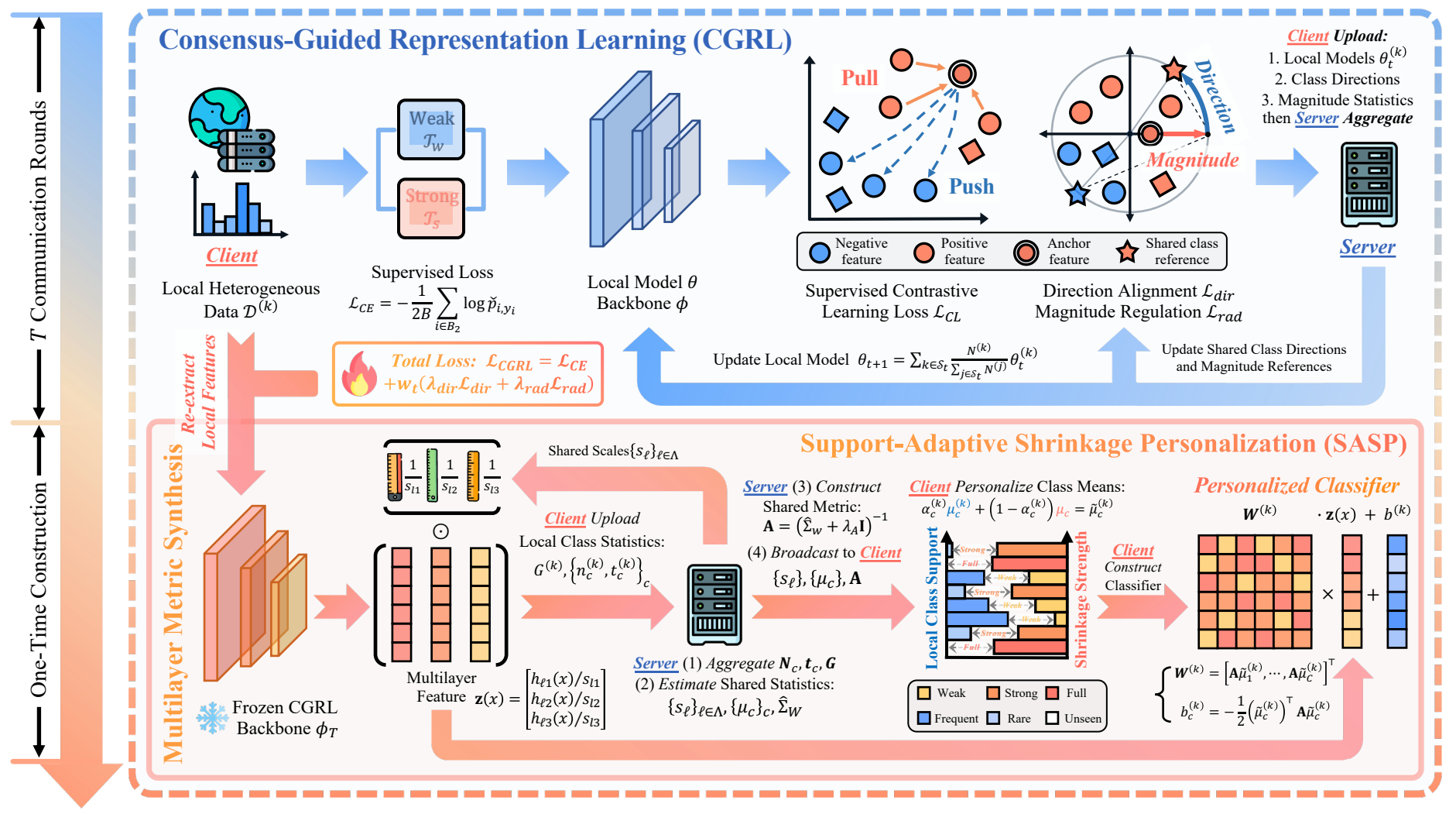

T Communication Rounds
One-Time Construction
Consensus-Guided Representation Learning (CGRL)
Client
Local Heterogeneous Data $\mathcal{D}^{(k)}$
Weak $\mathcal{T}_w$
Strong $\mathcal{T}_s$
Supervised Loss
$\mathcal{L}_{CE} = -\frac{1}{2B}\sum_{i\in B_2}\log \breve{p}_{i,y_i}$
Local Model θ
Backbone ϕ
Pull
Push
Negative feature
Positive feature
Anchor feature
Shared class reference
Supervised Contrastive Learning Loss $\mathcal{L}_{CL}$
Direction
Magnitude
Direction Alignment $\mathcal{L}_{dir}$
Magnitude Regulation $\mathcal{L}_{rad}$
Client Upload:
1. Local Models $\theta_t^{(k)}$
2. Class Directions
3. Magnitude Statistics
then Server Aggregate
Server
Total Loss: $\mathcal{L}_{CGRL} = \mathcal{L}_{CE} + w_t(\lambda_{dir}\mathcal{L}_{dir} + \lambda_{rad}\mathcal{L}_{rad})$
Update Local Model $\theta_{t+1} = \sum_{k\in\mathcal{S}_t}\frac{N^{(k)}}{\sum_{j\in\mathcal{S}_t}N^{(j)}}\theta_t^{(k)}$
Update Shared Class Directions and Magnitude References
Re-extract Local Features
Multilayer Metric Synthesis
Frozen CGRL Backbone ϕ_T
Multilayer Feature $\mathbf{z}(x) = [h_{\ell_1}(x)/s_{l1}; h_{\ell_2}(x)/s_{l2}; h_{\ell_3}(x)/s_{l3}]$
Support-Adaptive Shrinkage Personalization (SASP)
Shared Scales $\{s_\ell\}_{\ell\in\Lambda}$
Client Upload
Local Class Statistics:
$G^{(k)}, \{n_c^{(k)}, t_c^{(k)}\}_c$
Server (1) Aggregate $\mathbf{N}_c, \mathbf{t}_c, \mathbf{G}$
(2) Estimate Shared Statistics:
$\{s_\ell\}_{\ell\in\Lambda}, \{\mu_c\}_c, \hat{\Sigma}_W$
Server (3) Construct Shared Metric:
$\mathbf{A} = (\hat{\Sigma}_w + \lambda_A \mathbf{I})^{-1}$
(4) Broadcast to Client
$\{s_\ell\}, \{\mu_c\}, \mathbf{A}$
Client Personalize Class Means:
$\alpha_c^{(k)}\mu_c^{(k)} + (1-\alpha_c^{(k)})\mu_c = \tilde{\mu}_c^{(k)}$
Local Class Support
Shrinkage Strength
Weak
Strong
Full
Frequent
Rare
Unseen
Client Construct Classifier
Personalized Classifier
$\mathbf{W}^{(k)} \cdot \mathbf{z}(x) + b^{(k)}$
$\mathbf{W}^{(k)} = [\mathbf{A}\tilde{\mu}_1^{(k)}, \cdots, \mathbf{A}\tilde{\mu}_C^{(k)}]^\top$
$b_c^{(k)} = -\frac{1}{2}(\tilde{\mu}_c^{(k)})^\top \mathbf{A}\tilde{\mu}_c^{(k)}$